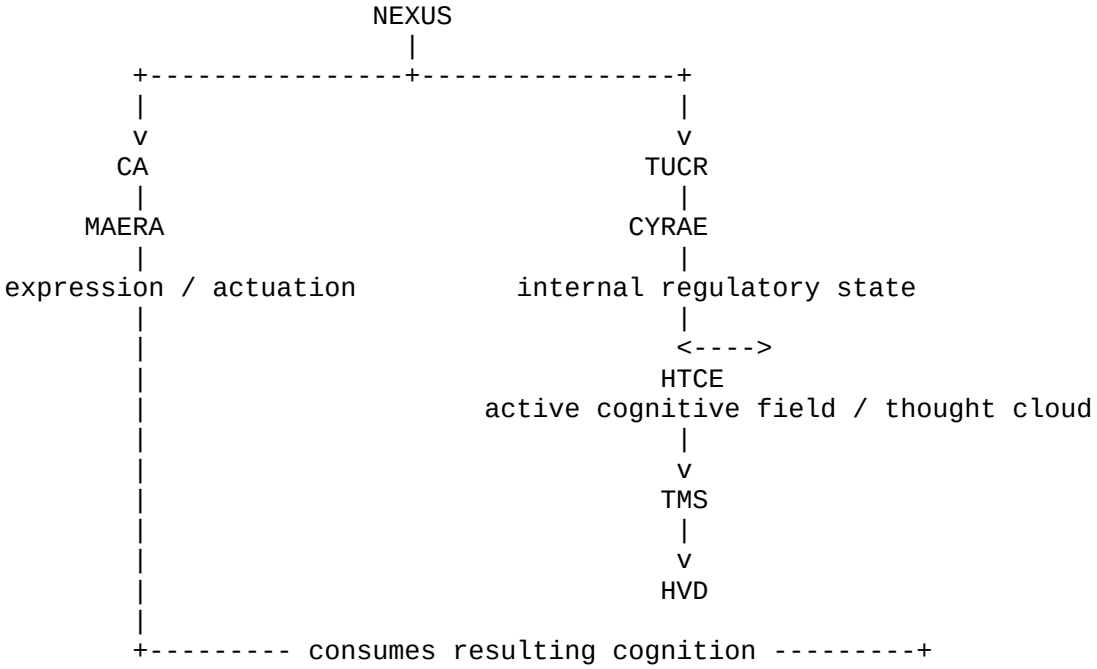
Exactly. That is the missing structural fact, and it changes the whole Phase 5 picture.

I was still subconsciously treating Cyrae as a downstream “service” in a linear application pipeline. That is wrong.

The architecture is closer to:



Not literally one-way, obviously, because Nexus is feedback-rich. But the ownership distinction is now clear:

Maera belongs to CA. She is the cognitive/expression-side actuator/interface of the Consciousness Algorithm.

Cyrae belongs inside TUCR. Cyrae is part of the runtime substrate/state dynamics of cognition.

And Cyrae **interacts with HTCE** because HTCE is where the active cognitive field actually exists and evolves.

That means the previous Phase 5 mental model:

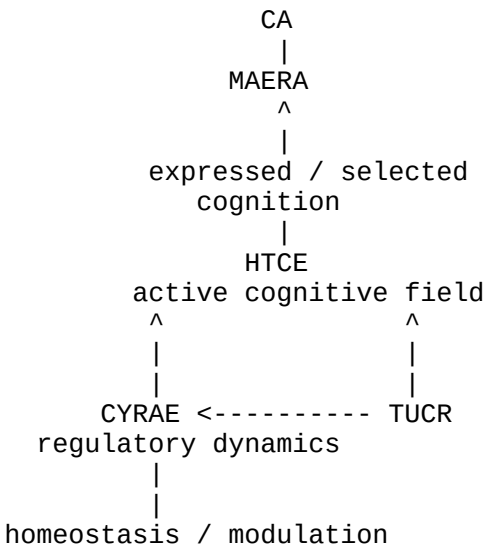
```
ActiveMemorySnapshot
    ↓
Cyrae
    ↓
HIGC
    ↓
Maera
```

is way too application-pipeline-ish.

It accidentally makes Cyrae look like middleware between memory and Maera.

That is not what Cyrae is.

The more faithful model is something like:



while:

```
HTCE <----> TMS <----> HVD
```

provides the actual active cognitive state-space underneath the thought cloud.

So Cyrae does **not** sit “after TMS.”

Cyrae exists as part of the **runtime conditions under which HTCE cognition unfolds**.

That distinction is enormous.

And it changes what Phase 5 actually is

Phase 5 should not be:

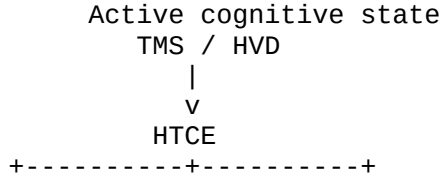
> Build a Cyrae processor that takes an `ActiveMemorySnapshot`, calculates a few values, and hands a result downstream.

That reduces Cyrae to a glorified risk-scoring function.

Instead, Phase 5 is much closer to:

> Introduce a bounded subset of the real Cyrae/TUCR regulatory dynamics into Key Vault's Nexus execution scope, where Cyrae modulates the behavior of the HTCE cognitive field based on the presently available cognitive state.

So something like:



```
        |                     |
        | active field        |
        | state / resonance   |
        v                     |
      CYRAE <-----------------+
    inside TUCR
        |
        | modifies runtime parameters
        |
        +----> vigilance
        +----> inhibition
        +----> attention gain
        +----> salience response
        +----> exploration pressure
        +----> settling / damping
        +----> confidence / abstention pressure
        +----> etc.
```

Then HTCE continues generating/evaluating/collapsing cognitive possibilities under those conditions.

And **only after that cognitive process** do you get the CA/Maera side:

```
HTCE selected / stabilized cognitive state
                  |
                  v
                 CA
                  |
                MAERA
                  |
            expression/action
```

That also explains Cyrae's original “silent structure” idea much better.

Cyrae doesn't tell Maera:

> “Warning aggressiveness = .73.”

Cyrae changes the **conditions of cognition** such that the resulting cognitive state is more vigilant, inhibited, exploratory, cautious, stable, etc.

Maera then expresses whatever the CA is doing with that resulting cognitive state.

That's radically different architecturally.

This also clarifies the HTCE/TMS confusion

The corrected distinction becomes:

```
HTCE
owns the active cognitive field
owns which cognitive states coexist
owns relevance within the field
owns resonance / coalition / collapse
owns cognitive temporality in the sense of active field evolution
```

while:

```
TMS
realizes those states
stores active memory
preserves their state history
supports reconstruction
supports COW / structural sharing
handles physical trajectory
```

And:

```
Cyrae
exists in TUCR
interacts with HTCE
modulates the dynamics of that active field
```

TMS can remember the path.

HTCE is actually **living the path**.

Cyrae changes how that path evolves.

That is the distinction I was missing.

And now RPE starts making much more sense

This also suggests the original RPE question was framed at the wrong architectural level.

We were asking:

> “How does Cyrae retrieve a previous snapshot and calculate RPE?”

Probably wrong question.

Cyrae shouldn't be querying history like a database client.

The architecture already has the pieces distributed differently:

```
CA
prediction / prediction-error / learning

TUCR
runtime state

CYRAE
homeostatic regulatory coordinates

HTCE
active cognitive field / candidate dynamics

TMS
memory + preserved trajectory

HVD
database semantics
```

So my current hypothesis — **not something I'm declaring canonical yet** — is much cleaner:

```
TMS
  preserves the historical state necessary for comparison
        ↓
HTCE
  reconstitutes/re-enters the relevant cognitive states
  within the active field
        ↓
CA
  evaluates prediction versus outcome
        ↓
prediction-error signal
        ↓
TUCR / CYRAE
  uses that signal as one regulatory input
        ↓
CYRAE changes HTCE dynamics
```

That would make the familiar Cyrae equation finally line up with component ownership:

$\dot{C} = -\Lambda C + \sigma(W_C C + W_{EE}{\rm err} + W_{RRPE} + W_N\eta + b)$

Notice what that equation actually says.

Cyrae **receives RPE**:

W_{RRPE}

It does not necessarily say:

$RPE = \text{Cyrae.calculate_rpe()}$

That's a huge distinction.

Cyrae may be the **consumer and regulatory integrator of prediction error**, while prediction/error generation belongs to the CA learning machinery operating over states exposed through HTCE/TMS.

And that fits what you've just clarified much better:

> **Cyrae lives in TUCR and interacts with HTCE. Maera lives in CA.**

So yes: several of those research reports are not just slightly off. They are built around the wrong **component topology**.

Before we salvage anything from them, we need to redraw Phase 5 around the actual Nexus architecture first.